

Foundations of Decision tree

[Decision tree]

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The basic interpretation in this situation is that the company prefers B's risk and payoffs under realistic risk preference coefficients (greater than \$400K—in that range of risk aversion, the company would need to model a third strategy, "Neither A nor B"). Another example, commonly used in operations research courses, is the distribution of lifeguards on beaches (a.k.a. the "Life's a Beach" example). The example describes two beaches with lifeguards to be distributed on each beach. There is maximum budget B that can be distributed among the two beaches (in total), and using a marginal returns table, analysts can decide how many lifeguards to allocate to each beach.

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Overview A decision tree is a flowchart-like structure in which each internal node represents a test on an attribute (e.g. whether a coin flip comes up heads or tails), each branch represents the outcome of the test, and each leaf node represents a class label (decision taken after computing all attributes). The paths from root to leaf represent classification rules. In decision analysis, a decision tree and the closely related influence diagram are used as a visual and analytical decision support tool, where the expected values (or expected utility) of competing alternatives are calculated. A decision tree consists of three types of nodes:

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The tree using information gain has the same results when using the phi function when calculating the accuracy. When we classify the samples based on the model using information gain we get one true positive, one false positive, zero false negatives, and four true negatives. For the model using the phi function we get two true positives, zero false positives, one false negative, and three true negatives. The next step is to evaluate the effectiveness of the decision tree using some key metrics that will be discussed in the evaluating a decision tree section below. The metrics that will be discussed below can help determine the next steps to be taken when optimizing the decision tree.

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Decision nodes – typically represented by squares
Chance nodes – typically represented by circles
End nodes – typically represented by triangles
Decision trees are commonly used in operations research and operations management. If, in practice, decisions have to be taken online with no recall under incomplete knowledge, a decision tree should be paralleled by a probability model as a best choice model or online selection model algorithm. Another use of decision trees is as a descriptive means for calculating conditional probabilities. Decision trees, influence diagrams, utility functions, and other decision analysis tools and methods are taught to undergraduate students in schools of business, health economics, and public health, and are examples of operations research or management science methods. These tools are also used to predict decisions of householders in normal and emergency scenarios.

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Optimizing a decision tree A few things should be considered when improving the accuracy of the decision tree classifier. The following are some possible optimizations to consider when looking to make sure the decision tree model produced makes the correct decision or classification. Note that these things are not the only things to consider but only some.

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The choice of node-splitting functions The node splitting function used can have an impact on improving the accuracy of the decision tree. For example, using the information-gain function may yield better results than using the phi function. The phi function is known as a measure of "goodness" of a candidate split at a node in the decision tree. The information gain function is known as a measure of the "reduction in entropy". In the following, we will build two decision trees. One decision tree will be built using the phi function to split the nodes and one decision tree will be built using the information gain function to split the nodes. The main advantages and disadvantages of information gain and phi function

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if condition1 and condition2 and condition3 then outcome. Decision rules can be generated by constructing association rules with the target variable on the right. They can also denote temporal or causal relations.

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Decision rules The decision tree can be linearized into decision rules, where the outcome is the contents of the leaf node, and the conditions along the path form a conjunction in the if clause. In general, the rules have the form:

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Evaluating a decision tree It is important to know the measurements used to evaluate decision trees. The main metrics used are accuracy, sensitivity, specificity, precision, miss rate, false discovery rate, and false omission rate. All these measurements are derived from the number of true positives, false positives, True negatives, and false negatives obtained when running a set of samples through the decision tree classification model. Also, a confusion matrix can be made to display these results. All these main metrics tell something different about the strengths and weaknesses of the classification model built based on your decision tree. For example, a low sensitivity with high specificity could indicate the classification model built from the decision tree does not do well identifying cancer samples over non-cancer samples. Let us take the confusion matrix below.

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This is the phi function formula. The phi function is maximized when the chosen feature splits the samples in a way that produces homogenous splits and have around the same number of samples in each split.

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We will set D , which is the depth of the decision tree we are building, to three ($D = 3$). We also have the following data set of cancer and non-cancer samples and the mutation features that the samples either have or do not have. If a sample has a feature mutation then the sample is positive for that mutation, and it will be represented by one. If a sample does not have a feature mutation then the sample is negative for that mutation, and it will be represented by zero. To summarize, C stands for cancer and NC stands for non-cancer. The letter M stands for mutation, and if a sample has a particular mutation it will show up in the table as a one and otherwise zero.